12TH CHEMISTRY · CHAPTER 24 · SOLVED EXERCISE

Organic Nitrogen Compounds

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Long Questions) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS:

I. Primary, secondary and tertiary amine are regarded as:

(a) Tautomers (b) Metamers
(c) Functional group isomers (d) Position isomers

Answer: (c) Functional group isomers — Amines of the same molecular formula (e.g. C_3H_9N) have different functional groups: $-NH_2$, >NH and >N–.

II. General formula for secondary amines is:

(a) RNH_2 (b) RNHR
(c) R_3N (d) $RCONH_2$

Answer: (b) RNHR — Two alkyl or aryl groups are attached to the NH group.

III. Which one of the following bases is a strongest one?

(a) Ammonia (b) Ethyl amine
(c) Phenyl amine (d) Diethyl amine

Answer: (d) Diethyl amine — It has two electron-donating ethyl groups, which make the lone pair on nitrogen most available.

Diethylamine is not discussed in the chapter; the answer follows from the textbook rule that electron-donating alkyl groups on nitrogen increase basicity.

IV. Acid hydrolysis if nitriles produces:

(a) Alcohol (b) Ketone
(c) Carboxylic acid (d) Ester

Answer: (c) Carboxylic acid — E.g. $CH_3CN + 2H_2O + HCl \rightarrow CH_3COOH + NH_4Cl$.

V. phenyl amine areacts with aqueous Br to form:

(a) Phenol (b) 2,4,6-Tribromophenylamine
(c) An azo compound (d) None of these

Answer: (b) 2,4,6-Tribromophenylamine — A white precipitate is formed; both ortho positions and the para position are substituted.

VI. Benzenediazonium chloride reacts with phenol in NaOH to form:

(a) Phenyl hydrozine (b) Nitro compound
(c) Azo compound (d) Alcohol

Answer: (c) Azo compound — Coupling gives p-hydroxyazobenzene, which contains the –N=N– group.

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VII. What is the structure of an azo compound:

(a) R_1-N=N-R_2
(b) R_1-NH-R_2
(c) R-O-R
(d) R-C≡N

Answer: (a) R_1-N=N-R_2 — Azo compounds contain the –N=N– bond.

VIII. When diazonium salt is heated with water, the gas evolved is:

(a) Nitrogen (b) Ammonia
(c) Nitric oxide (d) Nitrogen dioxide

Answer: (a) Nitrogen — $C_6H_5N_2^+Cl^- + H_2O \rightarrow C_6H_5OH + N_2 + HCl$.

IX. An aqueous solution of glycine is neutral because of the formation of:

(a) Carbanion (b) Carbocation
(c) Zwitterion (d) Free radical

Answer: (c) Zwitterion — The proton moves from the carboxyl group to the amino group, giving a dipolar ion (internal salt).

X. Which amines are produced by the reduction of amides?

(a) Primary (b) Secondary
(c) Tertiary (d) None of these

Answer: (a) Primary — Amides are reduced to primary amines with $LiAlH_4$.

MCQ IV ("if nitriles"), V ("areacts with aqueous Br") and VI ("Phenyl hydrozine") are reproduced as printed.

Q2. SHORT ANSWER QUESTIONS:

a) Give two preparations of phenyl amine

i) Ammonolysis of chlorobenzene: ammonolysis is a reaction in which an ammonia molecule is broken down to react with another compound.

$$C_6H_5–Cl + 2NH_3 \rightarrow C_6H_5–NH_2 + NH_4Cl\ (Cu_2O,\ 250\ °C,\ 50\ atm)$$

ii) Reduction of nitrobenzene:

$$C_6H_5–NO_2 + 6[H] \rightarrow C_6H_5–NH_2 + 2H_2O\ (Sn + conc.\ HCl)$$

b) Give two preparations of secondary amines.

i) Alkylation of primary amines: when a halogenoalkane reacts with a primary amine in the presence of ethanol under pressure, a secondary amine is formed.

$$CH_3–CH_2–Br + CH_3–NH_2 \rightarrow CH_3–CH_2–NH–CH_3 + HBr\ (ethanol)$$

ii) Reduction of N-substituted amides:

$$CH_3–CO–NH–CH_3 + 4[H] \rightarrow CH_3–CH_2–NH–CH_3 + H_2O\ (H_2\ with\ Ni/Pt,\ or\ LiAlH_4)$$

c) What is the chromophore in azo dyes? What is its function in dyes?

The azo group (Ph–N=N–Ph) in azo dyes is called a **chromophore**. This group is responsible for the intense colour of the dyes, ranging from yellow and orange to red, blue and green, depending on what other groups are attached to the benzene rings. Different food colours often contain azo dyes.

d) Why amides are less basic than ammonia.

In an amide the lone pair on nitrogen is delocalised with the oxygen of the carbonyl group (resonance). This reduces the availability of the electrons on nitrogen for accepting a proton. In ammonia the lone pair is fully available. Hence amides are much weaker bases and form neutral solutions in water.

e) What is zwitter ion?

An α-amino acid exists as a dipolar ion called a **zwitterion**, which has a positive and a negative end within the same molecule. It is formed when the proton goes from the carboxyl group to the amino group. The dipolar structure is also called the internal salt.

$$H_2N–CH(R)–COOH \rightarrow H_3N^{+}–CH(R)–COO^{-}$$

f) Differentiate between primary, secondary & tertiary amines.

Primary amine	Secondary amine	Tertiary amine
One alkyl or aryl group is attached to nitrogen.	Two alkyl or aryl groups are attached to nitrogen.	Three alkyl or aryl groups are attached to nitrogen.
Contains the $–NH_2$ group; $R–NH_2$.	Contains the >NH group; R_2NH.	Has no N–H bond; R_3N.
E.g. ethanamine	E.g. N-methylethanamine	E.g. N,N-dimethylmethanamine
With nitrous acid: diazonium salt, nitrogen gas.	With nitrous acid: yellow oily N-nitrosoamine.	With nitrous acid: soluble nitrite salt, no visible change.

g) Write names & structures of four isomeric amines having the molecular formula C_3H_9N.

#	Name	Structure	Type
1	propan-1-amine	$CH_3–CH_2–CH_2–NH_2$	Primary
2	propan-2-amine	$CH_3–CH(NH_2)–CH_3$	Primary
3	N-methylethanamine	$CH_3–CH_2–NH–CH_3$	Secondary
4	N,N-dimethylmethanamine	$(CH_3)_3N$	Tertiary

h) Convert ethanamine into methanamine.

The carbon chain has to be shortened by one carbon atom; this is done in the last step by the Hofmann reaction.

$$CH_3–CH_2–NH_2 + HNO_2 \rightarrow CH_3–CH_2–OH + N_2 + H_2O\ (NaNO_2/HCl)$$
$$CH_3–CH_2–OH + 2[O] \rightarrow CH_3–COOH + H_2O\ (K_2Cr_2O_7 / H_2SO_4, \text{reflux})$$
$$CH_3–COOH + SOCl_2 \rightarrow CH_3–COCl + SO_2 + HCl$$
$$CH_3–COCl + NH_3 \rightarrow CH_3–CO–NH_2 + HCl$$
$$CH_3–CO–NH_2 + Br_2 + 4NaOH \rightarrow CH_3–NH_2 + 2NaBr + Na_2CO_3 + 2H_2O$$

The first step (primary aliphatic amine + nitrous acid → alcohol + nitrogen) is standard chemistry: the textbook says only that the alkyl diazonium salt is highly unstable and that nitrogen gas is produced. The other steps are textbook reactions (Chapters 23 and 24).

i) How will you synthesize ethanal from ethanenitrile?

Ethanenitrile is hydrolysed to ethanoic acid, which is reduced to ethanol; ethanol is then oxidised and the ethanal is distilled off as it forms.

$$CH_3–CN + 2H_2O + HCl \rightarrow CH_3–COOH + NH_4Cl\ (\text{heat})$$
$$CH_3–COOH + 4[H] \rightarrow CH_3–CH_2–OH + H_2O\ (LiAlH_4)$$
$$CH_3–CH_2–OH + [O] \rightarrow CH_3–CHO + H_2O\ (K_2Cr_2O_7 / H_2SO_4, \text{distil})$$

j) Define peptide bond. How many peptide bonds are there in a tripeptide?

A peptide bond (peptide link) is the amide bond, –CO–NH–, formed between two amino acid molecules when the –COOH group of one reacts with the $–NH_2$ group of the other with the elimination of water. A tripeptide is formed from three amino acids, so it contains **two** peptide bonds.

k) Give all the tripeptides that could possibly be synthesized from glycine (Gly), alanine (Ala).

Each of the three positions can be glycine or alanine, so 2 × 2 × 2 = **8** tripeptides are possible (written from the amino end to the carboxyl end):

Both amino acids present (6)	Only one amino acid (2)
Gly–Gly–Ala, Gly–Ala–Gly, Ala–Gly–Gly	Gly–Gly–Gly
Gly–Ala–Ala, Ala–Gly–Ala, Ala–Ala–Gly	Ala–Ala–Ala

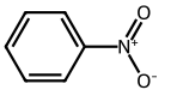
Q3. CONSTRUCTED RESPONSE QUESTIONS:

Complete the following reaction sequence by giving the structures of the compounds.

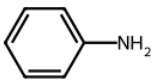
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a) 1, 3, 5-tribromobenzene from benzene. benzene → (HNO_3/H_2SO_4) (A) → (Sn/HCl) (B) → (Br_2 (excess)/$FeBr_3$) (C) → ($NaNO_2$/HCl) (D) → (H_3PO_2, $-N_2$) (E)

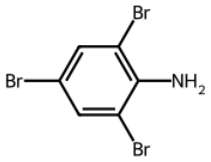
Benzene is nitrated to nitrobenzene (A), which is reduced to phenylamine (B). The $-NH_2$ group directs bromine to both ortho positions and the para position, giving 2,4,6-tribromophenylamine (C). Diazotization gives the diazonium salt (D), and H_3PO_2 replaces the diazonium group by hydrogen with loss of N_2, leaving 1,3,5-tribromobenzene (E).



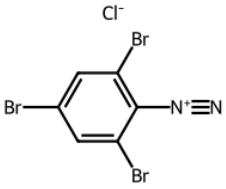
A: nitrobenzene



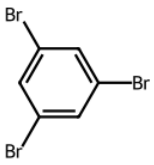
B: phenylamine



C: 2,4,6-tribromophenylamine



D: 2,4,6-tribromobenzenediazonium chloride

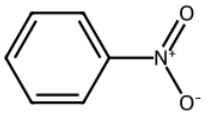


E: 1,3,5-tribromobenzene

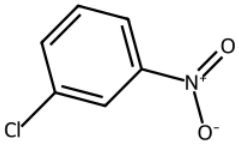
The replacement of the diazonium group by hydrogen with H_3PO_2 is not described in the chapter; it is given by the scheme ($-N_2$) and by the title of the question.

b) 1, 3, 5-Tribromobenzene from benzene. benzene ring → (HNO_3/H_2SO_4) (A) → ($Cl_2/FeCl_3$) (B) → (Sn/HCl) (C) → ($NaNO_2$/HCl) (D) → (H_2O, $-N_2$) (E)

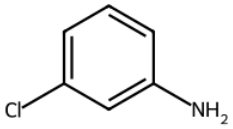
Benzene is nitrated to nitrobenzene (A). The $-NO_2$ group is meta directing, so chlorination gives 3-chloronitrobenzene (B). Reduction gives 3-chlorophenylamine (C), diazotization gives 3-chlorobenzenediazonium chloride (D), and warming with water replaces the diazonium group by –OH, giving 3-chlorophenol (E).



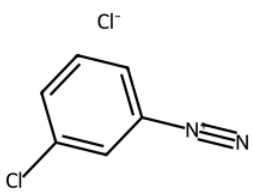
A: nitrobenzene



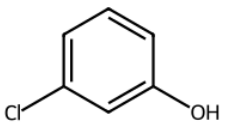
B: 3-chloronitrobenzene



C: 3-chlorophenylamine



D: 3-chlorobenzenediazonium chloride

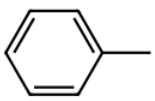


E: 3-chlorophenol

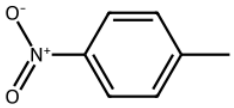
The heading of part b) repeats "1, 3, 5-Tribromobenzene from benzene", but the reagents of this scheme (Cl_2/$FeCl_3$, then H_2O) lead to 3-chlorophenol, not to 1,3,5-tribromobenzene. The scheme is solved as printed.

c) Keep in view the scheme, devise a reaction sequence to prepare 4-Hydroxybenzoic acid from benzene.

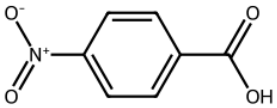
Step	Reagents and conditions	Product
1	CH_3Cl, $AlCl_3$ (Friedel–Crafts alkylation)	methylbenzene
2	conc. HNO_3 / conc. H_2SO_4 ($-CH_3$ is ortho-para directing; para isomer separated)	4-nitromethylbenzene
3	$KMnO_4$, dil. H_2SO_4, heat (side chain oxidised)	4-nitrobenzoic acid
4	Sn / conc. HCl (reduction)	4-aminobenzoic acid
5	$NaNO_2$ / HCl, below 10 °C (diazotization)	diazonium salt
6	H_2O, warm ($-N_2$)	4-hydroxybenzoic acid



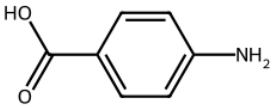
1: methylbenzene



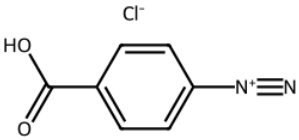
2: 4–nitromethylbenzene



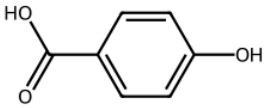
3: 4–nitrobenzoic acid



4: 4–aminobenzoic acid



5: diazonium salt



6: 4–hydroxybenzoic acid

LONG QUESTIONS

Q4. Describe the relative basicity of aqueous ammonia, ethyl amine and phenyl amine.

Amines which are soluble in water form weakly alkaline solutions, just as ammonia does, due to partial reaction with the solvent forming OH^- ions.

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^- \ (pK_b = 4.75)$$
$$CH_3\text{–}CH_2\text{–}NH_2 + H_2O \rightleftharpoons CH_3\text{–}CH_2\text{–}NH_3^+ + OH^- \ (pK_b = 3.29)$$
$$C_6H_5\text{–}NH_2 + H_2O \rightleftharpoons C_6H_5\text{–}NH_3^+ + OH^- \ (pK_b = 9.38)$$

Sr.No.	Amine	Formula	K_b	pK_b
1	Ammonia	NH_3	1.8×10^{-5}	4.75
2	Ethanamine	$CH_3\text{–}CH_2\text{–}NH_2$	5.1×10^{-4}	3.29
3	Phenylamine (aniline)	$C_6H_5\text{–}NH_2$	4.2×10^{-10}	9.38

The basicity depends on the availability of the lone pair of electrons on the nitrogen atom. The larger the K_b (or the smaller the pK_b), the stronger the base.

Ethanamine is a stronger base than ammonia. The ethyl group is electron-donating; it pushes electron density towards the nitrogen atom, so the lone pair is more available to accept a proton. Likewise, alkyl amines in general are stronger bases than ammonia.

Ammonia is a stronger base than phenylamine. In phenylamine the lone pair of electrons on the nitrogen atom is delocalised over the benzene ring. The nitrogen atom is planar, with its lone pair in a p orbital, which overlaps with the delocalised π electrons of the benzene ring. The lone pair is therefore less available to accept a proton.

Basic strength: ethanamine > ammonia > phenylamine

Q5. Give two preparation methods of each of the following. i) Primary amines ii) Secondary amines

i) Primary amines

i) From halogenoalkanes and ammonia: primary amines are prepared by the nucleophilic substitution of halogenoalkanes with ammonia. Excess of ethanolic ammonia under pressure is used to avoid the formation of secondary amines or a mixture of products.

$$CH_3\text{–}CH_2\text{–}Br + NH_3 \text{ (excess)} \rightarrow CH_3\text{–}CH_2\text{–}NH_2 + HBr \text{ (ethanol, high pressure)}$$

ii) Reduction of alkanenitriles with H_2/Ni or $LiAlH_4$:

$$CH_3\text{–}CH_2\text{–}C{\equiv}N + 4[H] \rightarrow CH_3\text{–}CH_2\text{–}CH_2\text{–}NH_2 \ (LiAlH_4)$$

ii) Secondary amines

i) Alkylation of primary amines: when a halogenoalkane reacts with a primary amine in the presence of ethanol under pressure, a secondary amine is formed.

$$CH_3–CH_2–Br + CH_3–NH_2 \rightarrow CH_3–CH_2–NH–CH_3 + HBr \text{ (ethanol)}$$

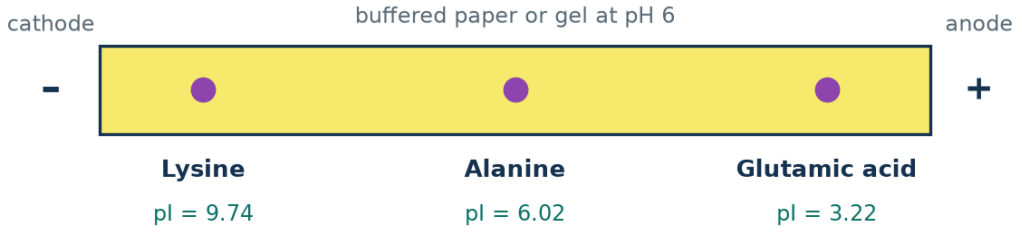
ii) Reduction of N-substituted amides:

$$CH_3–CO–NH–CH_3 + 4[H] \rightarrow CH_3–CH_2–NH–CH_3 + H_2O \ (H_2 \text{ with Ni/Pt, or } LiAlH_4)$$

Q6. Write a note on electrophoresis.

Electrophoresis is used to separate amino acids in a mixture on the basis of their charge. A few drops of the mixture are applied to a filter paper or gel placed in a buffered solution between two electrodes. When an electric field is applied, the amino acids separate according to their different pI values:

- If the pI of an amino acid is **greater** than the pH of the solution, it exists mainly in a form with a positive charge and migrates towards the **cathode**.
- If the pI is **lower** than the pH of the solution, it exists mainly in a form with a negative charge and migrates towards the **anode**.
- The greater the difference between pI and pH, the faster it migrates.
- If two amino acids have very similar pI values (such as glycine and leucine), the one with the larger molecular weight moves more slowly, because the charge has to carry a greater mass.



Separation of amino acids at pH 6 by electrophoresis

Amino acids are colourless, so a detection technique is needed to see the location of the spots. **Ninhydrin** is used as a locating agent; it reacts with amino acids to produce a purple product (with H_2O, CO_2 and RCHO as by-products).

Electrophoresis cannot be used to separate large quantities of amino acids. It is only an analytical method for determining the number of amino acids in a mixture. To actually separate an entire mixture, other laboratory techniques are used, such as column chromatography.

Q7. Explain the formation of azo compounds from phenylamine and illustrate the preparation of a dye using this?

Step 1 – Diazotization:

Phenylamine, being a primary amine, reacts with nitrous acid to form a phenyldiazonium salt. This reaction is called **diazotization**. Organic compounds having the diazonium ion, $–N^+{\equiv}N$ (diazo group), as functional group are called diazo

compounds. Nitrous acid is unstable and is prepared in situ from sodium nitrite and HCl. The reaction mixture is kept below 10 °C using ice, because diazonium salts are unstable and decompose easily, giving off nitrogen gas, even at room temperature.

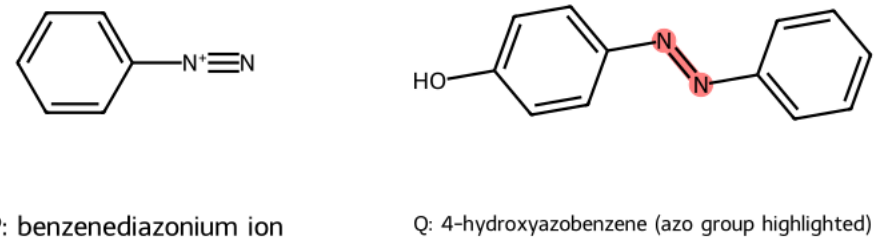
$$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$$

$$C_6H_5\text{–}NH_2 + HNO_2 + HCl \rightarrow C_6H_5\text{–}N_2^+Cl^- + 2H_2O \text{ (less than 10 °C)}$$

Step 2 – Coupling:

The reactions of aryl diazonium salts with strongly activated aromatic compounds, such as phenols and amines, are called **coupling reactions**. The products, which contain an –N=N– bond, are called **azo compounds**. In the coupling reaction the positively charged diazonium ion acts as an electrophile and substitutes into the benzene ring of phenol at the para position.

$$C_6H_5\text{–}N_2^+Cl^- + C_6H_5\text{–}OH \rightarrow C_6H_5\text{–}N{=}N\text{–}C_6H_4\text{–}OH + HCl \text{ (NaOH)}$$



P: benzenediazonium ion Q: 4-hydroxyazobenzene (azo group highlighted)

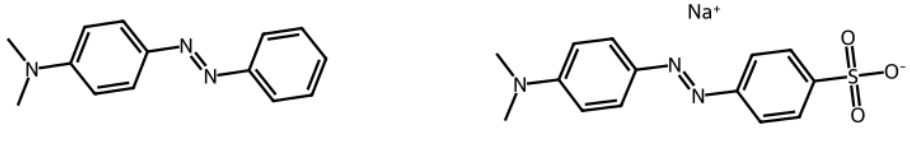
Benzenediazonium ion and p-hydroxyazobenzene (orange solid); the azo group is highlighted

Mechanism: the terminal nitrogen of the diazonium ion (the electrophile) is attacked by the electron-rich para carbon of phenol, giving an intermediate that carries a positive charge; this then loses H^+ from that carbon, the ring becomes aromatic again and p-hydroxyazobenzene is formed.

Preparation of a dye:

Coupling reactions are used to form different types of azo dyes. **p-Dimethylaminoazobenzene** is called butter yellow; it was previously used in food, but because of its carcinogenic nature its use in food is now forbidden. It is made by coupling benzenediazonium chloride with N,N-dimethylphenylamine. **Methyl orange**, an acid–base indicator, is prepared from diazotized sulfanilic acid and N,N-dimethylphenylamine.

$$C_6H_5\text{–}N_2^+Cl^- + C_6H_5\text{–}N(CH_3)_2 \rightarrow C_6H_5\text{–}N{=}N\text{–}C_6H_4\text{–}N(CH_3)_2 + HCl \text{ } (CH_3COOH)$$



p-dimethylaminoazobenzene (butter yellow) methyl orange

The azo group (Ph–N=N–Ph) in azo dyes is called a **chromophore**. This group is responsible for the intense colour of the dyes, ranging from yellow and orange to red, blue and green, depending on what other groups are attached to the benzene rings. Different food colours often contain azo dyes.

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 24.1

a) Give IUPAC names of the following compounds. i) CH_3–CH(CH_3)–CH_2–CH_2–CH_2–NH_2 ii) CH_3–CH_2–CH(NH)–CH_3 iii) CH_3–CH_2–CH(CH_2–CH_3)–CH(CH_2–CH_3)–NH–CH_2–CH_3

i) 4-methylpentan-1-amine

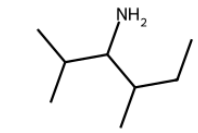
ii) butan-2-amine

iii) N,4-diethylhexan-3-amine (a secondary amine; the longest chain through the carbon bonded to nitrogen has six carbon atoms)

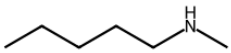
In part ii) the group is printed as "NH", and the textbook itself adds "(note: likely intended to be NH_2 for 2-butanamine)". It is named here as butan-2-amine.

b) Draw a structure corresponding to each amine. i) 2,4 dimethylhexan -3-amine ii) N-methylpentan -1- amine iii) 2-methylcyclohexan-1-amine iv) N, N -dimethylethanamine

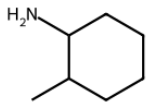
	Name	Condensed structure
i)	2,4-dimethylhexan-3-amine	CH_3–CH(CH_3)–CH(NH_2)–CH(CH_3)–CH_2–CH_3
ii)	N-methylpentan-1-amine	CH_3–CH_2–CH_2–CH_2–CH_2–NH–CH_3
iii)	2-methylcyclohexan-1-amine	cyclohexane ring with –NH_2 on C-1 and –CH_3 on C-2
iv)	N,N-dimethylethanamine	CH_3–CH_2–N(CH_3)$_2$



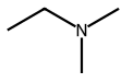
i) 2,4-dimethylhexan-3-amine



ii) N-methylpentan-1-amine



iii) 2-methylcyclohexan-1-amine



iv) N,N-dimethylethanamine

Quick Check 24.2

a) The "feel good factor" in chocolate has been identified as 2-phenylethanamine, $C_6H_5CH_2CH_2NH_2$. Suggest a synthesis of this compound from chloromethylbenzene, $C_6H_5CH_2Cl$.

Step 1: heat chloromethylbenzene with KCN in ethanol; the chlorine is replaced by –CN and the chain becomes one carbon longer.

$$C_6H_5–CH_2–Cl + KCN \rightarrow C_6H_5–CH_2–CN + KCl \text{ (ethanol, heat)}$$

Step 2: reduce the nitrile with $LiAlH_4$ (or H_2/Ni).

$$C_6H_5–CH_2–CN + 4[H] \rightarrow C_6H_5–CH_2–CH_2–NH_2 \ (LiAlH_4)$$

b) What nitro compound, nitrile and amide are reduced to yield each of the following amine compound? I) For the primary amine: CH_3–CH(CH_3)–CH_2–NH_2 (determine the structures of nitro compound, nitrile, and amide that would produce this amine upon reduction) ii) For the secondary amine: CH_3–CH_2–CH_2–CH_2–NH–CH_3

I) 2-Methylpropan-1-amine is obtained by the reduction of:

Class	Compound	Structure
Nitro compound	2-methyl-1-nitropropane	$(CH_3)_2CH–CH_2–NO_2$
Nitrile	2-methylpropanenitrile	$(CH_3)_2CH–C\equiv N$
Amide	2-methylpropanamide	$(CH_3)_2CH–CO–NH_2$

ii) N-Methylbutan-1-amine is a secondary amine. It is obtained by the reduction of an N-substituted amide: N-methylbutanamide, $CH_3–CH_2–CH_2–CO–NH–CH_3$ (or N-butylmethanamide, H–CO–NH–CH_2–CH_2–CH_2–CH_3). The reduction of a nitro compound or a nitrile gives only a primary amine (–NH_2), so neither can give this secondary amine directly.

"Nh_2" and "Ch_3" are printed in the formulae of this Quick Check; they are –NH_2 and –CH_3.

Quick Check 24.3

a) Phenylamine is about a million times less basic than cyclohexylamine. Give reason. (phenylamine $K_b = 4.2 \times 10^{-10}$; cyclohexylamine $K_b = 3.3 \times 10^{-4}$)

In phenylamine the lone pair on nitrogen is delocalised over the benzene ring (it is in a p orbital which overlaps with the π electrons of the ring), so it is much less available to accept a proton. In cyclohexylamine the ring is saturated: there is no delocalisation, and the alkyl ring is electron-donating, which makes the lone pair more available.

b) Which of ethanamine and propan-2-amine is more basic? Explain.

Propan-2-amine is more basic. Basicity depends on the availability of the lone pair on nitrogen, and electron-donating alkyl groups increase it. In propan-2-amine the carbon bonded to nitrogen carries two methyl groups, so more electron density is pushed towards nitrogen than by the ethyl group of ethanamine.

c) Show the reaction of propan-2-amine with HCl. Name the salt formed.

$$(CH_3)_2CH–NH_2 + HCl \rightarrow (CH_3)_2CH–NH_3^+Cl^-$$

The salt is propan-2-ammonium chloride (isopropylammonium chloride).

d) Which acid reacts with N,N-dimethylammine in fish when it is treated with vinegar? Write a chemical equation to show this reaction.

Vinegar contains **ethanoic acid** (acetic acid). It reacts with the amine of fish, N,N-dimethylmethanamine (trimethylamine), to form a salt which is less smelly.

$$(CH_3)_3N + CH_3–COOH \rightarrow (CH_3)_3NH^+\ CH_3–COO^-$$

Quick Check 24.4

Azo dyes are the coloured compounds which are prepared by the reaction of diazonium salt of phenylamine and some aryl compound having activated benzene ring. i) How will you prepare a dye from phenylamine and 2-methylphenol. ii) Give equations stepwise along with reaction conditions.

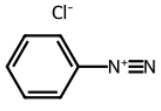
i) Phenylamine is first diazotized with nitrous acid below 10 °C. The cold benzenediazonium chloride solution is then added to 2-methylphenol dissolved in sodium hydroxide solution (mild alkaline conditions). The diazonium ion couples at the position para to the –OH group, giving an azo dye.

ii) Equations:

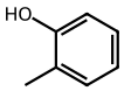
$$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$$

$$C_6H_5–NH_2 + HNO_2 + HCl \rightarrow C_6H_5–N_2^+Cl^- + 2H_2O \text{ (less than 10 °C)}$$

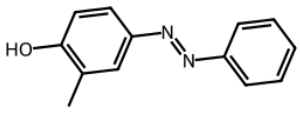
$$C_6H_5–N_2^+Cl^- + CH_3–C_6H_4–OH \rightarrow C_6H_5–N{=}N–C_6H_3(CH_3)–OH + HCl\ (NaOH)$$



benzenediazonium chloride



2-methylphenol

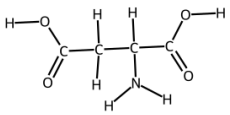


azo dye (4-hydroxy-3-methylazobenzene)

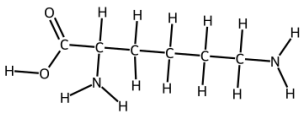
Quick Check 24.5

The R groups in the amino acids aspartic acid and lysine are ($-CH_2-COOH$) and ($-CH_2-(CH_2)_3-NH_2$) respectively. a) Draw the displayed formulae of these amino acid. b) Draw the structures of their Zwitter ions. c) Draw the structures of their ions in acidic conditions. d) Draw the structures of their ions in basic conditions. Discuss with what we mean by the isoelectric point of these amino acids.

a) Displayed formulae:



aspartic acid



lysine

	Aspartic acid	Lysine
Formula	$H_2N–CH(CH_2–COOH)–COOH$	$H_2N–CH[(CH_2)_4–NH_2]–COOH$
b) Zwitterion	$H_3N^+–CH(CH_2–COOH)–COO^-$	$H_3N^+–CH[(CH_2)_4–NH_2]–COO^-$
c) Acidic conditions	$H_3N^+–CH(CH_2–COOH)–COOH$	$H_3N^+–CH[(CH_2)_4–NH_3^+]–COOH$
d) Basic conditions	$H_2N–CH(CH_2–COO^-)–COO^-$	$H_2N–CH[(CH_2)_4–NH_2]–COO^-$

Isoelectric point: it is the pH at which the amino acid has no overall charge in aqueous solution, i.e. the concentration of its zwitterion is at a maximum, so it does not move in an electric field. Each amino acid has its own pI. Aspartic acid has an extra acidic –COOH group, so its isoelectric point lies at a low (acidic) pH; lysine has an extra basic $–NH_2$ group, so its isoelectric point lies at a high (basic) pH (9.74).

Zwitterions are shown in the general textbook form; the low pI of aspartic acid is general knowledge (the book gives the pI of lysine only).